GEON CHOI

Ph.D. Student @ KAIST AI
Internship Availability: June 2026 – February 2027

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RESEARCH INTERESTS

My research focuses on **Foundation Models** for **Multimodal Learning**, with an emphasis on **Vision-Language Models (VLMs)** and **Large Language Models (LLMs)** applied to **AI for Healthcare**. I am particularly interested in advancing **Computer Vision** and **Generative Modeling** to build large-scale, real-world systems that bridge cutting-edge foundation model research with high-impact medical applications.

PUBLICATIONS

CHIL 2026
Lunguage: A Benchmark for Structured and Sequential Chest X-ray Interpretation
Jong Hak Moon, Geon Choi, Paloma Rabaey, Min Gwan Kim, Hyuk Gi Hong, Jung-Oh Lee, Hangyul Yoon, Eun Woo Doe, Jiyoun Kim, Harshita Sharma, Daniel C. Castro, Javier AlvarezValle, Edward Choi
Conference on Health, Inference, and Learning, 2026

Contribution: Developed an LLM-based framework for structuring radiology reports.
Links: [arXiv] [Code]

CVPR 2026
Instruction-Guided Lesion Segmentation for Chest X-rays with Automatically Generated Large-Scale Dataset
Geon Choi*, Hangyul Yoon*, Hyunju Shin, Hyunki Park, Sang Hoon Seo, Eunho Yang, Edward Choi
(*: Equal Contribution)
IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026

Contribution: Co-led the project; designed and implemented the automatic large-scale dataset generation framework.
Links: [arXiv] [Code]

NeurIPS 2025
Spotlight
CXReasonBench: A Benchmark for Evaluating Structured Diagnostic Reasoning in Chest X-rays
Hyungyung Lee, Geon Choi, Jung-Oh Lee, Hangyul Yoon, Hyuk Gi Hong, Edward Choi
Conference on Neural Information Processing Systems (Datasets and Benchmarks Track), 2025

Contribution: Built the information-extraction pipeline for X-ray image quality assessment sub-tasks (inclusion, rotation).
Links: [arXiv] [Code]

Under Review
CheXpercept: A Benchmark for Evaluating Expert-Level Lesion Perception in Chest X-rays
Geon Choi, Hangyul Yoon, Nalee Kim, Jeong Yun Jang, Hyunju Shin, Hyunki Park, Sang Hoon Seo, Edward Choi

Contribution: Led the project end-to-end; designed and built the benchmark and conducted all model evaluations.

EDUCATION

2/2026 - Present
Korea Advanced Institute of Science and Technology (KAIST) — **Seongnam, South Korea**
Ph.D., Artificial Intelligence (Advisor: Prof. Edward Choi)

2/2024 - 2/2026
Korea Advanced Institute of Science and Technology (KAIST) — **Seongnam, South Korea**
M.S., Artificial Intelligence (Advisor: Prof. Edward Choi)

2/2020 - 2/2024
Pohang University of Science and Technology (POSTECH) — **Pohang, South Korea**
B.S., Electrical Engineering
GPA: 4.04/4.3 (Summa Cum Laude)
Ranked 2nd in the department of electrical engineering.

3/2018 - 2/2020
Jeonbuk Science High School — **Iksan, South Korea**
Early Graduation
Ranked 3rd in class, earning the POSTECH Presidential Fellowship.

EXPERIENCE

6/2023 – 8/2023
KAIST AI Research Internship (KAIRI) — **Daejeon, South Korea**
- BioImaging, Signal Processing and Learning Lab, KAIST (Advisor: Prof. Jongchul Ye)
- Research Topic: Generative Models for Solving Inverse Problems in Medical Imaging

1/2023 – 6/2023
Undergraduate Research Internship — **Pohang, South Korea**
- Bio Optics and Acoustics Lab, POSTECH (Advisor: Prof. Chulhong Kim)
- Research Topic: Computer-Aided Diagnosis System for Breast Ultrasonography

HONORS AND AWARDS

2024 - Present **KAIST Scholarship** **KAIST**
Tuition Scholarship

2023 **KEC Scholarship** **KEC Foundation**
Merit Scholarship for Outstanding EE Student ($2k)

2022 - 2023 **National Science & Technology Scholarship** **KOSAF**
Full Tuition Scholarship

2020 - 2021 **POSTECH Jigok Scholarship** **POSTECH**
Full Tuition Scholarship

2020 - 2023 **POSTECH Presidential Fellowship** **POSTECH**
Merit Fellowship for Exceptional Freshman ($10k)

TEACHING

2022 **Course Assistant for Digital System Design (EECE273)** **POSTECH**
Student Mentor

SERVICES

2024 - 2025 **KoSAIM Summer School Hands-on Session** **KoSAIM**
Instructor: taught chest X-ray classification with CNN for medical AI researchers and medical students in South Korea.

LANGUAGES

English - professional working proficiency
Korean - native

SKILLS

Programming Languages: Python, C
Deep Learning: PyTorch, HuggingFace Transformers, DeepSpeed, Accelerate
Foundation Models: LLM / VLM training and inference, diffusion models, vLLM
Medical Imaging: pydicom, OpenCV, scikit-image
Tools & Infrastructure: Git, Linux, TensorBoard